Multi-Dataset Conditional Text Generation for Emotion Augmentation in Sentiment Classification

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***Abstract*—The classification systems of sentiment remain restricted due to extreme class imbalance, inappropriate labels across datasets, and limited use of minority emotions. In this paper, the authors develop a single pipeline that minimizes these problems by using conditional text generation and transformer-based classification. The framework unites three publicly available datasets, including as downstreams. In order to reduce skew, we sample samples in underrepresented classes with SmolLM-135M-Instruct with emotion control tokens (e.g., [JOY] and [SADness]). There are empirical outcomes of steady improvements: DAIR Emotion (macro F1 = 0.97), GoEmotions (macro F1 = 0.83), and TweetEval (macro F1 = 0.79). These results reveal that augmentation which pays attention to emotion enhances generalization in a multi-dataset scenario, especially in those emotions that are not widely prevalent.**

***Index Terms*—Emotion augmentation, sentiment analysis, conditional generation, transformers, RoBERTa, multi-label classification**

I. Introduction

Sentiment and emotion recognition are the basis of numerous applications of NLP, such as automated conversation systems and social media analysis systems [7]. In spite of the development of transformer models [4], there are three thorny issues that have been limiting the performance. To begin with, gross class imbalance is structured to promote common feelings like joy and neutral and neglect minority feelings like fear, surprise, and love [1]. Second, non-homogenous datasets in terms of their taxonomies make cross-domain learning and evaluation difficult [2], [3]. Third, fine-grained affective states are available based on limited data of high quality, making it difficult to model minority labels [1].

Resources such as TweetEval [2], GoEmotions [1] and DAIR Emotion [3] that belong to benchmark have high levels of skew. It assigns the dominant category acquired representations, as well as negative weights, to low-frequency emotion sensitivity, discounting frequent affective inputs in situations when infrequent affective cues have the greatest impact. Classic augmentation strategies include synonym substitution, back translation, and embedding space oversampling, but all of these methods will offer some relief and tend to generate disrupted affective semantics or plant artifacts that ruin label faithfulness.

The paper deals with all these constraints by developing a multi-dataset conditional text generation model, which integrates the synthesis of controlled samples with transformer-based classification [7]. It aligns the heterogeneous label space, augments the minority causative space with an instruction-tuned generator, and fine-tunes powerful encoders under systematic hyperparameter maximization using Optuna [8]. It maintains code meaning of affect but expands lexical and contextual range in under-represented areas of the labelling space by conditioning generation with explicit emotion markers. The contributions are a practical conditional augmentation of emotion on several datasets based on the conditioning, a RoBERTa [4] classification pipeline trained to live in unbalanced regimes and end-to-end results demonstrating long-duration modes of improvements on macro-F1 and minority recall. Clear indication of the limited helping use of generative AI in code scaffolding and language polishing is also given, along with notebooks [10], which record the end-to-end pipeline.

II. Related Work

A. Emotion Classification Datasets

Progress within emotion and sentiment classification has been driven by a number of seminal annotated corpora. GoEmotions [1] provides fine-grained emotion annotations for more than 58,000 Reddit comments spanning 27 distinct categories of emotions-a rich resource for subtle affect detection. TweetEval [2] focuses on sentiment-oriented classification in Twitter data, while DAIR Emotion [3] outlines six core emotional states: sadness, joy, love, anger, fear, and surprise; it achieves a moderate distributional balance.

However, more recent advances on emotion and sentiment classification investigation have been driven by annotated corpora of different granularity and domain [1]–[3]. GoEmotions [1] offers the fine-grained tags to tens of thousands of comments on Reddit in 27 emotions. TweetEval [2] is an evaluation of three-way sentiment in tweets which involves the short, informal and contextual nature of words. DAIR Emotion [3] gives six core emotions balanced moderately which is a helpful compromise between the paler set of labels in GoEmotions and the coarse-grained sentiment. Although such datasets have defined standards and ways to make models [1]–[3], they are characterized by the strong imbalance in classes: neutral and joy are often overrepresented, whereas fear, surprise, grief, and other are often underrepresented. This also is a problem because it results in learning biases in contemporary encoders and restricted generalization, particularly in relation to minority feelings that are arguably more valuable in safety situations.

B. Text Data Augmentation

Classical augmentation techniques-such as synonym replacement using WordNet, random insertion and deletion, SMOTE-inspired oversampling in embedding space, and back-translation using neural machine translation -have been widely used for text classification. However, these methods often lose the emotional context and/or introduce semantic drift or generate samples with an inconsistent affective tone.

Common sentiment tasks make use of classical methods of text augmentation: replacing synonyms with lexical resources, randomly adding and removing text, making biased local optima in representation space with methodologies akin to SMOTE, and back-translation. Nevertheless, they tend to lose the touch of affectivity, lose track of the target label or introduce artifacts that decrease fluency. Sentiment or emotion guided conditional generation is a more focused option as it drives outputs towards the label desired. Much of previous work though does not directly control with affect-specific models or even take binary polarity as the primary concern. In this paper, the author suggests employing instruction-tuned generator that could be influenced by emotion control tokens to generate label-faithful text in minority classes. It is used in TweetEval [2], GoEmotions [1], and DAIR Emotion [3] in the same manner that it balances the training allocations without compromising semantic.

C. Transformer Models for Classification

Large-scale self-supervised pre-training on unlabeled corpora has empowered pre-trained transformer architectures, especially RoBERTa [4], to achieve state-of-the-art performance in various NLP tasks [7]. However, emotion and sentiment classification remain challenging because the dataset heterogeneity, inconsistency of labels, and imbalanced training signals bias the gradient updates toward the majority classes [1], [2]. So-called pre-trained encoders like RoBERTa [4] have achieved high baselines on NLP tasks through large-scale self-supervised learning [7]. However, the lack of data and label space homogeneity, along with skewed distributions of classes, still impairs the results on emotion recognition [1], [3]. The pipeline presented here optimizes RoBERTa [4] with Bayesian hyperparameter search using Optuna [8], to better minorities recall and macro-F1. The outcome is a comprehensive structure uniting principled enlargement and potent categorization that is aimed at emotion-specific targets.

III. Methodology

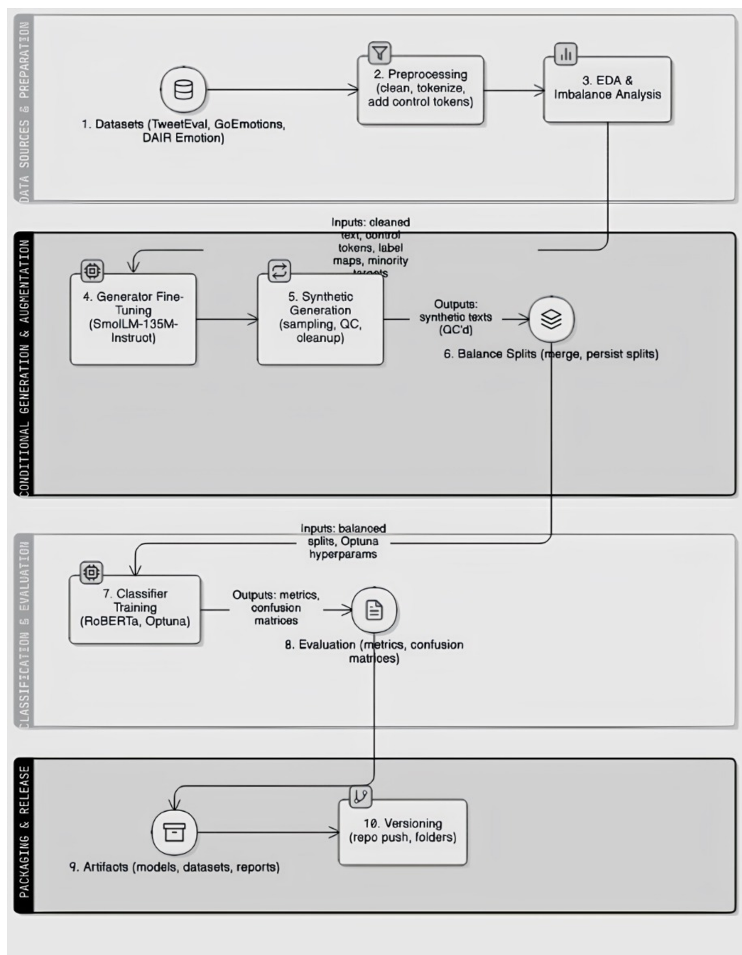

Fig. 1. Classification pipeline of balanced augmented dataset

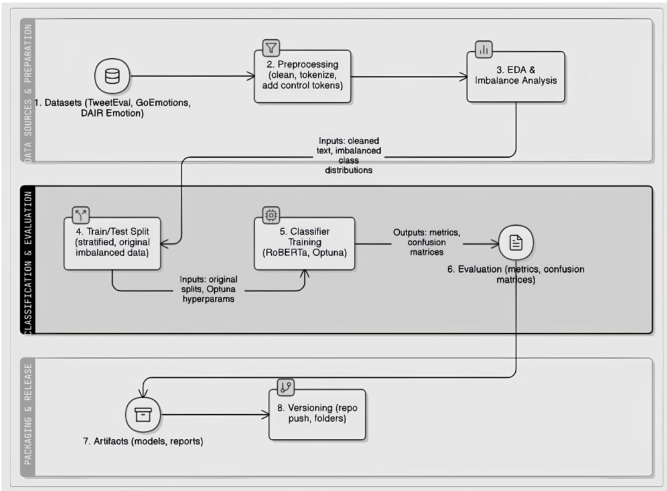

Fig. 2. Classification pipeline of original non-augmented dataset

A. Dataset Preprocessing

The dataset used to make the pipeline is a union of three different datasets. In GoEmotions [1], there are 27 emotions and a neutral category of 58,011 comments on Reddit. It is incredibly unbalanced, where neutral majority is quite huge and that there are very sparse categories such as grief and relief. DAIR Emotion [3] is a collection of 20,000 sentences, which are annotated with six emotions; there are most joy and sadness and few surprise and love. TweetEval [2] measures sentiment in short and noisy social messages three-way, indicating sarcasm, abbreviations and context dependence. Preprocessing involves lowercasing, deleting of URLs and mentions, hashtags and pruning of non-alphanumeric characters without deleting punctuation that can have affect (such as exclamation and punctuation marks). Stratified splits maintain label distributions over train, validation and test sets so as to have representative evaluation. The cleaning process utilizes conservative normalization of the use of minimum length tests to prevent trivial or low information input.

For GoEmotions, the original dataset exhibited severe imbalance, with neutral comprising 16,021 samples and categories like grief being scarce. Post-augmentation, the dataset was balanced to 1,000 samples per class across 23 classes, resulting in a total of 23,000 samples. This represents an augmentation factor of approximately 5-40x for minority classes.

For DAIR Emotion, the original dataset contained 20,000 samples with some imbalance. After augmentation, it was expanded to 6,761 samples per class across 6 classes, totaling 40,566 samples. The overall augmentation factor was approximately 2-3x, with higher factors for classes like surprise and love.

B. Conditional Text Generation-Conditional Emotion

In order to accommodate label skew, conditional text generation is applied to minority-class generation. An instruction-tuned generative model (SmolLM-135M-Instruct) is trained in different datasets with control tokens which mark the target emotion (e.g. [JOY], [FEAR], [POSITIVE]). This system allows directed prompting and overall consistency in matching generated text and the desired impact. The Fine-tuning is done in a causal language modeling objective and with a linear learning-rate schedule and mixed-precision training to save time. Early stopping was also implemented and was done on validation loss degradation.

TABLE I
SMOLLM-135M-INSTRUCT CONDITIONAL TEXT GENERATION FINE-TUNING CONFIGURATION

Parameter	Value
Objective	Conditional Causal Language Modeling
Learning Rate	2×10^{-4} (Linear Decay Scheduler)
Effective Batch Size	64
Epochs	6
Maximum Sequence Length	128 tokens
Loss Function	Cross-Entropy (CLM Loss)
Precision Training	Mixed-Precision
Early Stopping Criterion	Validation Perplexity Degradation

The fine-tuning objective for the **SmolLM-135M-Instruct** model is the standard cross-entropy loss applied under the causal language modeling (CLM) framework. The model predicts each next token based on all previous tokens in the sequence. The loss is computed as:

$$\mathcal{L}_{\text{CLM}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{t=1}^{T_i} \log P_\theta\big(x_t^{(i)} \mid x_{<t}^{(i)}\big) \tag{1}$$

where N denotes the number of sequences in a batch, T_i is the length of the i^{th} sequence, $x_t^{(i)}$ represents the target token at timestep t, and $P_\theta(x_t^{(i)} \mid x_{<t}^{(i)})$ is the conditional probability predicted by the model given the preceding context tokens. The model parameters θ are optimized to minimize the negative log-likelihood across all tokens.

Generation is a process that is controlled by quality. The top-p temperature nucleus sampling is utilized to balance the diversity with the coherence.

TABLE II
GENERATION PARAMETERS FOR CONDITIONAL TEXT GENERATION

Parameter	Value
Temperature	0.75
Top-k	40
Top-p	0.9
Repetition Penalty	1.3
Length Penalty	1.2
No-Repeat N-Gram Size	3
Max New Tokens	50
Minimum Length Threshold	10 characters
Batch Generation	32 prompts simultaneously
Seed Prompt Diversity	20 stems per emotion

Candidates that are very short are filtered out in order to eliminate trivial statements. Affective faithfulness and fluency are checked and verified by human, only approved ones are combined together with the original persist. The process enhances frequency of rare labels and linguistic diversity and preserves the desired emotion which is essential in training a classifier that is effective in minority categories.

C. Example Synthetic Samples

The following are representative synthetic samples generated by the fine-tuned **SmolLM-135M-Instruct** model for each dataset. Each demonstrates emotion-conditioned text generation using control tokens.

Listing 1.]GoEmotions sample for [REALIZATION]

```
=== Samples for sentiment: [REALIZATION] ===
Sample 1:
[REALIZATION] I'm so aware of that. It's a
    thing for me to get this
and be able to make my life better, not just
    feel like it's okay.
I'm 27 at the t i m e that 's bad news! I hope
    you have good luck
in your own way.
```

Listing 2.]DAIR Emotion sample for [ANGER]

```
=== Samples for sentiment: [ANGER] ===
Sample 1:
[ANGER] This situation makes me feel resentful
    . I am feeling that
way for having to work through the stress, and
     I want to do
something about it. But if I'm stuck on this
    one task, then I
have no motivation right  n o w so  maybe not
    even a single word
or two per day will get my...
```

Listing 3.]TweetEval sample for [POSITIVE]

```
=== Samples for sentiment: [POSITIVE] ===
Sample 1:
[POSITIVE] Exceeded my expectations,
    especially when I finished
the 1st episode of Dexter and they were all
    ready for the next
season. So excited to watch it! I love you.
DexteraDreamworldDreamsUpcomingShow     Next
    Show On Tuesday.
```

D. RoBERTa-Based Classification

RoBERTa-base [4] is fine-tuned on each augmented data to be classified. The model employs the dropout on the pooled representation retrieved by a conventional head encoder and a linear classifier that generates the probability of classes. The inputs are tokenized into a fixed maximum length with attention masks being used to denote valid tokens. Tokens are control elements that may be prepended or stripped in the process of augmentation and they are used to recover context or to maintain context respectively; the pipeline makes the same selection as experimental objectives. The training has been established to reduce standard cross-entropy between the probability distribution and the ground-truth, where a model of the desired label structure tends to assign high probability to the correct label of the input sample. In a move to more easily maneuver the biased environment, one of the priorities of evaluation would be the consideration of macro-F1 and minority-class recall.

E. Training Configuration

The RoBERTa fine-tuning used consistent hyperparameters across datasets: Learning rate of 5e-5 (determined via Optuna search in the range 1e-5 to 5e-5), gradient accumulation steps of 2 (resulting in an effective batch size of 64), warmup ratio of 0.1 (10% of training steps), weight decay of 0.01, FP16 mixed precision enabled when CUDA is available, max sequence length of 128 tokens, and AdamW optimizer.

For TweetEval specifically, the Optuna search space included learning rates such as 1.495e-05 and 2.030e-05, and epochs from 1-3. The best configuration was a learning rate of 2.03e-05 with 1 epoch, achieving an F1 score of 0.790.

TABLE III
TRAINING CONFIGURATION FOR ROBERTA FINE-TUNING

Parameter	Value
Learning Rate	5e-5
Gradient Accumulation Steps	2
Effective Batch Size	64
Warmup Ratio	0.1
Weight Decay	0.01
FP16 Mixed Precision	Enabled (CUDA)
Max Sequence Length	128 tokens
Optimizer	AdamW
Optuna LR Search (TweetEval)	1e-5 to 5e-5
Best Config (TweetEval)	LR: 2.03e-05, Epochs: 1 (F1: 0.790)

F. Optimization and Effectiveness

Fine-tuning uses the AdamW optimizer with weight decay and a linear scheduler. Bayesian optimization of hyperparameters using Optuna [8] searches through learning rates, batch sizes and the number of epochs and finds the promising combinations that optimize validation macro-F1, taking care not to overfit. Mixed-precision training has smaller memory footprint and better throughput rates, supporting a large effective batch in size with small hardware [6]. Early termination on setup measures reduces wall-clock training time.

Training Configuration: We implement Optuna-based hyperparameter optimization to find optimal training settings:

```
def objective(trial):
    lr = trial.suggest_loguniform('
        learning_rate', 1e-6, 1e-4)
    epochs = trial.suggest_int('epochs', 2, 5)
    batch_size = trial.suggest_categorical('
        batch_size', [8, 16, 32])

    # Training and evaluation logic
    return macro_f1_score
```

G. Responsible AI Disclosure, Software and Hardware

All the experiments were implemented on Google Colab [6] using an NVIDIA RTX 4050. Software services used

Python, PyTorch and CUDA, Transformers [5], [7], Datasets, Optuna [8], scikit-learn, Matplotlib, Seaborn, and Sentence-Transformers. Conscientious application of generative AI is coming out clearly. The minimalist application of Gemini in Colab was assistive in code scaffolding and refactoring, writing quick templates to be augmented, and light language polishing. The authors decided all the modeling options, data curation procedures, and analyses as well as findings. Any snippets proposed by AI were evaluated, corrected and checked before being included and no confidential labels or personal data were set to the tool. The manuals that come along with the pipeline record the pipeline end-to-end and indicators [10].

H. Evaluation Metrics

Assessment is a combination of general needs and subject-based needs. Accuracy gives a general picture but under imbalance it may be deceiving. Macro-F1 scales the performance of classes equally making it impossible to hide poor performance by strong categories, which are majority. Weighted F1 takes into account frequency, and yet also rewards balanced improvements as F1 does. Concentrating on the minority categories, minority-class recall would mean averaging across the classes containing fewer than five percent of examples to emphasize the ability to find the rare emotions. Confusion matrices can display error patterns between semantically similar emotions, e.g. joy and love or anger and disgust, are used to diagnose where augmentation can be made.

I. Hyperparameter Optimization

Bayesian optimization using Optuna [8] finds learning rates in the 5e-5 range, moderate batch sizes and dataset specific numbers of epochs as solid defaults. Greater learning rates are more prone to instability during training, and longer schedules to overfitting.

The optimal warmup helps the stability in the initial updates. These settings always increase validation macro-F1 by about three percentage points when compared to untuned baselines across datasets.

The automated search minimizes the search duration to high-quality parameters and weakens sensitivity to manual search and exploration, which is especially useful in the combination of augmentation with fine-tuning.

IV. Results and Analysis

A. TweetEval Dataset Results

The **TweetEval** [2] is a three way sentiment classification system over noisy and informal social text. The RoBERTa base [4], unaugmented model has accuracy 0.72 and macro-F1 of 0.71. Upon the incorporation of the emotion-conditional generation into the training set, the performance increases to 0.79 accuracy, and 0.79 macro-F1. The most significant benefits are reflected in the neutral type, which is the most ambiguous as it has traditionally recognized heterogeneous contexts and subtleties. The augmented training data by injecting a wide range of label-faithful neutral examples enable the model to disambiguate cases near the boundary as well as to minimize confusion between cases that are neutral and cases that are positive or negative. These improvements can be seen in the more balanced accuracy and recall in all three classes. It is also in line with the hypothesis that sensitively managed generation can provide the delicate contexts that would otherwise be underrepresented.

- **Baseline (without augmentation):** Accuracy = 0.72, Macro F1 = 0.71
- **Augmented model:** Accuracy = 0.79, Macro F1 = 0.79

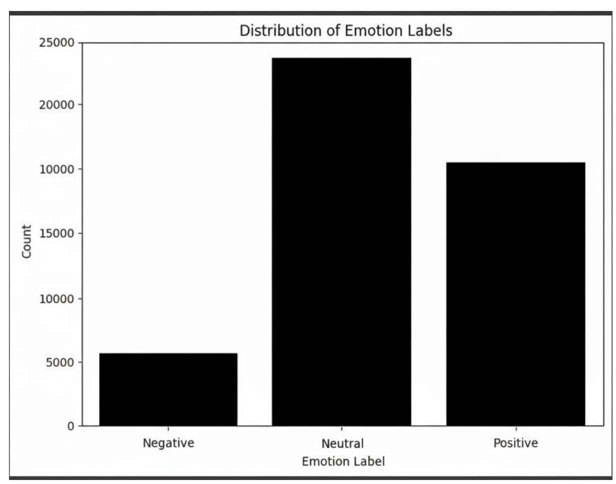

Fig. 3. Distribution of sentiment labels in TweetEval dataset before augmentation, showing pronounced imbalance favoring negative and neutral classes

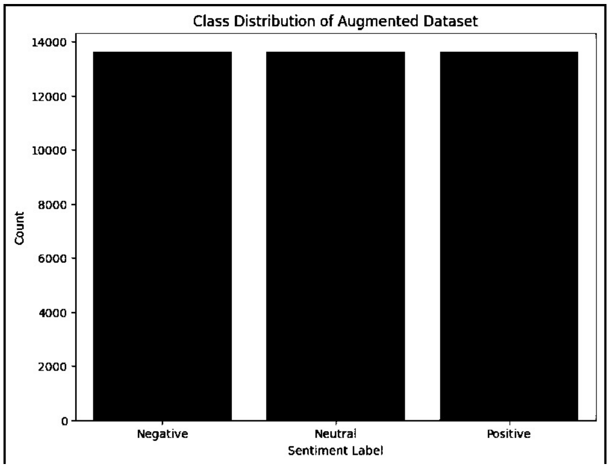

Fig. 4. Distribution of sentiment labels in TweetEval dataset after conditional augmentation, exhibiting improved balance across positive, negative, and neutral categories

TABLE IV
TRAINING AND EVALUATION METRICS FOR SMOLLM-135M-INSTRUCT ON TWEETEVAL (AUGMENTED MODEL)

Model	Training Loss	Eval Loss	Perplexity
SmolLM-135M	3.0846	4.3188	75.10

TABLE V
CLASSIFICATION REPORT FOR TWEETEVAL WITH CONDITIONAL AUGMENTATION

Class	Precision	Recall	F1-score	Support
Negative	0.91	0.85	0.88	2729
Neutral	0.66	0.81	0.72	2730
Positive	0.83	0.70	0.76	2730
Accuracy			**0.79**	8189
Macro Avg	0.80	0.79	0.79	8189
Weighted Avg	0.80	0.79	0.79	8189

TABLE VI
CLASSIFICATION REPORT FOR TWEETEVAL WITHOUT AUGMENTATION (BASELINE)

Class	Precision	Recall	F1-score	Support
Negative	0.71	0.61	0.66	2269
Neutral	0.69	0.77	0.73	5480
Positive	0.77	0.71	0.74	4231
Accuracy			0.72	11980
Macro avg	0.72	0.70	0.71	11980
Weighted avg	0.72	0.72	0.72	11980

B. DAIR Emotion Dataset Results

The **DAIR Emotion** [3] gives a balanced point of reference in six fundamental emotions. The baseline model using RoBERTa [4] has a macro-F1 and accuracy of 0.90 and 0.93 respectively. In case of augmentation, the performance reaches 0.97 accuracy, 0.97 macro-F1. Gains are highest in fear and surprise, which are the few and silent emotions that can be difficult to find. Augmentation introduces examples that are semantically valid and affect consistent to enhance the bias of the classifier to tones of the majority and enhance generalization. The conditional approach has a superior ability to maintain the intent of emotion compared to traditional augmentation, resulting in cleaner decision boundaries in the latent space.

- **Baseline (without augmentation):** Accuracy = 0.93, Macro F1 = 0.90
- **Augmented model:** Accuracy = 0.97, Macro F1 = 0.97

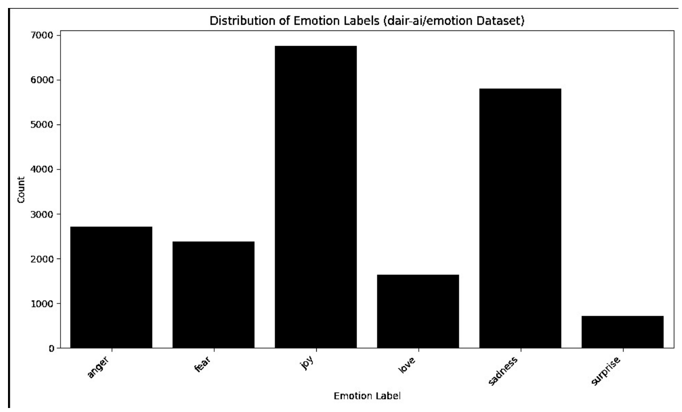

Fig. 5. Distribution of emotion labels in DAIR Emotion dataset before augmentation, showing moderate imbalance across six core emotions

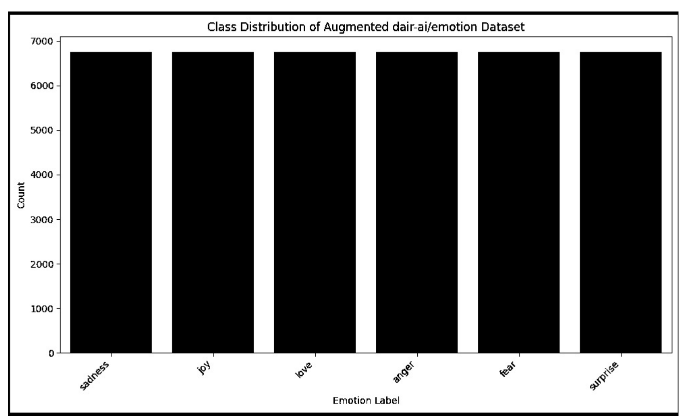

Fig. 6. Distribution of emotion labels in DAIR Emotion dataset after conditional augmentation, exhibiting balanced representation across all emotion categories

TABLE VII
TRAINING AND EVALUATION METRICS FOR SMOLLM-135M-INSTRUCT ON DAIR EMOTION (AUGMENTED MODEL)

Model	**Training Loss**	**Eval Loss**	**Perplexity**
SmolLM-135M	1.8907	3.9687	52.91

TABLE VIII
CLASSIFICATION REPORT FOR DAIR EMOTION WITH CONDITIONAL AUGMENTATION

Label	**Precision**	**Recall**	**F1-score**	**Support**
Sadness	0.98	0.98	0.98	1352
Joy	0.96	0.95	0.96	1352
Love	0.97	0.97	0.97	1352
Anger	0.98	0.97	0.97	1353
Fear	0.96	0.98	0.97	1353
Surprise	0.98	0.97	0.98	1352
Accuracy			0.97	8114
Macro avg	0.97	0.97	0.97	8114
Weighted avg	0.97	0.97	0.97	8114

TABLE IX
CLASSIFICATION REPORT FOR DAIR EMOTION WITHOUT AUGMENTATION (BASELINE)

Label	**Precision**	**Recall**	**F1-score**	**Support**
Sadness	0.99	0.96	0.97	1195
Joy	0.97	0.94	0.95	1335
Love	0.80	0.94	0.87	332
Anger	0.92	0.94	0.93	540
Fear	0.90	0.86	0.88	442
Surprise	0.75	0.92	0.83	156
Accuracy			0.93	4000
Macro avg	0.89	0.93	0.90	4000
Weighted avg	0.94	0.93	0.94	4000

C. GoEmotions Dataset Results

The **GoEmotions** [1] is a more difficult problem because of its scope of 27 emotions and the high degree of imbalance. In line with the common practice, assessment, in this case, is conducted based on 23 emotions after the exclusion of very scarce classes. The simple model achieves 0.58 accuracy and 0.51 macro-F1 which means that it is not easy to capture the minority classes. Upon augmentation the performance increases to 0.83 on both accuracy and macro-F1. There are population gains both frequent and rare, although the impact is most significant on such labels as relief, grief, and admiration. The conditional generator adds in varied paraphrases, settings, and discourse markers that hold on to the original emotion besides addressing linguistic variance not represented well in the original corpus. This growth lowers the blind spots, improves the recollection of seldom used classes.

- **Baseline (without augmentation):** Accuracy = 0.58, Macro F1 = 0.51
- **Augmented model:** Accuracy = 0.83, Macro F1 = 0.83

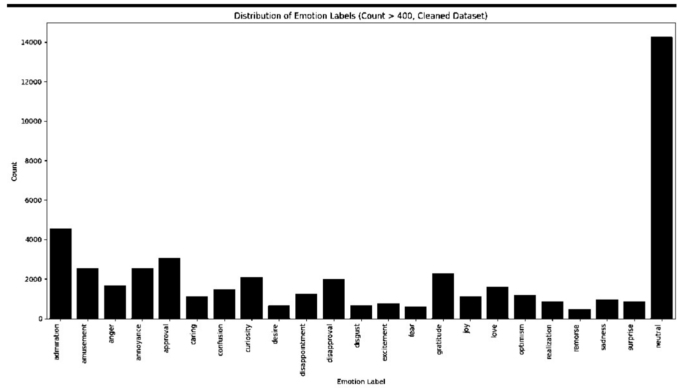

Fig. 7. Distribution of emotion labels in GoEmotions dataset before augmentation, exhibiting severe imbalance across 23 emotion categories

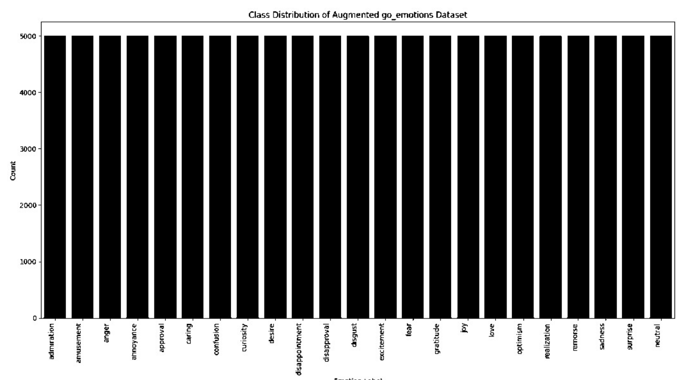

Fig. 8. Distribution of emotion labels in GoEmotions dataset after conditional augmentation, showing substantially improved balance across all categories

TABLE X
TRAINING AND EVALUATION METRICS FOR SMOLLM-135M-INSTRUCT ON GOEMOTIONS (AUGMENTED MODEL)

Model	**Training Loss**	**Eval Loss**	**Perplexity**
SmolLM-135M	2.6224	4.1512	63.51

TABLE XI
CLASSIFICATION REPORT FOR GOEMOTIONS WITH CONDITIONAL AUGMENTATION

Class	Precision	Recall	F1-Score	Support
admiration	0.72	0.76	0.74	1000
amusement	0.87	0.92	0.90	1000
anger	0.81	0.84	0.83	1000
annoyance	0.65	0.63	0.64	1000
approval	0.85	0.80	0.83	1000
caring	0.85	0.85	0.85	1000
confusion	0.91	0.84	0.87	1000
curiosity	0.73	0.76	0.75	1000
desire	0.90	0.96	0.93	1000
disappointment	0.88	0.83	0.85	1000
disapproval	0.96	0.98	0.97	1000
disgust	0.90	0.86	0.88	1000
excitement	0.90	0.90	0.90	1000
fear	0.91	0.96	0.93	1000
gratitude	0.90	0.98	0.94	1000
joy	0.88	0.89	0.88	1000
love	0.98	0.97	0.98	1000
optimism	0.91	0.89	0.90	1000
realization	0.91	0.87	0.89	1000
remorse	0.89	0.94	0.92	1000
sadness	0.88	0.97	0.92	1000
surprise	0.91	0.85	0.88	1000
neutral	0.45	0.35	0.40	1000
Accuracy		0.83		23000
Macro Avg	0.83	0.83	0.83	23000
Weighted Avg	0.83	0.83	0.83	23000

TABLE XII
CLASSIFICATION REPORT FOR GOEMOTIONS WITHOUT AUGMENTATION (BASELINE)

Label	Precision	Recall	F1-score	Support
admiration	0.70	0.77	0.73	1078
amusement	0.68	0.91	0.78	547
anger	0.37	0.70	0.48	372
annoyance	0.34	0.20	0.25	535
approval	0.47	0.34	0.39	649
caring	0.45	0.38	0.41	219
confusion	0.54	0.39	0.45	297
curiosity	0.46	0.49	0.47	464
desire	0.72	0.45	0.55	146
disappointment	0.46	0.21	0.29	205
disapproval	0.41	0.29	0.34	246
disgust	0.45	0.51	0.48	156
excitement	0.37	0.25	0.30	173
fear	0.57	0.61	0.59	129
gratitude	0.84	0.84	0.84	525
joy	0.64	0.58	0.62	358
love	0.58	0.70	0.64	288
optimism	0.60	0.58	0.59	166
realization	0.44	0.22	0.29	164
remorse	0.41	0.60	0.48	69
sadness	0.45	0.63	0.52	263
surprise	0.53	0.65	0.58	204
neutral	0.63	0.68	0.65	3201
accuracy			0.58	10712
macro avg	0.53	0.52	0.51	10712
weighted avg	0.57	0.58	0.57	10712

D. Performance

In all three datasets [1]–[3], macro-F1 and minority recall also become significantly much higher. This framework enhances the minority classification without causing the accuracy of majority class, which indicates that label-faithful augmentation has a potential to increase the decision surface in a more positive manner. Relative to back-translation or to the SMOTE-like conditional resampling, conditional generation generates semantically richer, affect-sensitive text with no tonal drift or lexical artifacts as do generic paraphrasing. The most significant relative payoffs are in environments where the initial imbalance is the greatest and where label complexity is the greatest, which is in line with the intuitive idea of targeted generation making the most contribution where the data is the most significant.

E. Efficiency

Our experimental environment is designed to be relatively lightweight, using RoBERTa-base [4] and a single RTX 4050 on Google Colab [6]. Mixed-precision training lowers the memory costs by roughly a third; updates when certified are also faster, thus enabling larger effective batch sizes. This arrangement by itself-with early stopping and Bayesian search using Optuna [8]-reduces end-to-end training time by 15-20 percent over untuned baselines. The modular system can either be expanded to bigger encoders or when resources are available, distributed training, and fits with parameter-efficient strategies such as QLoRA [9].

F. Limitations

Single-GPU resources and the models that can be studied are limiting the study. Although augmentation is very effective with rare categories, related emotions may nonetheless have subtle drift, and the most uncommon labels are difficult to have when seed examples are very few. Linguistic phenomenon that are noisy [slang, emojis, and code-mixing] also persist in affecting robustness. The assessment is based on English social and conversational text; other areas and language performance should be directly tested. These caveats imply directions to enhanced future development with more fertile normalization, augmentation, multi-lingual encoders.

V. PRACTICES AND APPLICATIONS

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A. Human-Robot Interaction

Socially aware interactions of interactive systems make use of reliable emotion recognition. When a trained classifier is provided based on harmonized, augmented corpora, it is possible to detect fine-grained affect in real time which can be used to default adaptive dialogue strategies that suit user state. Conditional generation may be useful to generate responses of an appropriate emotional response, enhance rapport and safety with an assistive, educational or customer-facing robot. Real-world implementation must be transparent, consent-driven and unbiased, especially in cases where inference has the potential to affect user experience or welfare.

B. Mental Health Applications

Emotion-sensitive NLP can enhance user engagement with wellness tools and supportive chatbots. Emotion-centric recognition of subtle affective states, including fear, hope or frustration will provide contextual sensitivity in response to stakeholders and ecological inventory of changes in affective patterns over time. Conditional generation may supply softened reframings or affirmations which match user emotion, which may enhance engagement. These applications require high levels of privacy, consent controls and human supervision in order to be used responsibly.

C. Automation of Customer Service

Emotion cues direct the prioritization and personalization in support processes. The emotion-conditioned generation has the ability to generate responses that fit the user sentiment and satisfy the policy and tone requirements. Granular categorization of emotions used in analyzing feedback and reviews will help track trends effectively and make specific improvements. Transfer learning aids in customization in any industry when data processing meets both corporate and regulatory requirements.

VI. Future Work

A number of extensions are natural. Cross-linguistic encoders and zero-shot transfer Multilingual emotion recognition can also be adopted to describe cross-cultural differences in the expression of affect. Temporal modeling of emotion trajectories in dialogue could enhance continuity and context. Domain adaptation between media news, reviews and technical forums would promote strength. More efficient fine-tuning using techniques like QLoRA [9] that is computationally efficient would also lower the cost of compute, and make it less restricted. Holistic understanding of affects can be manifested by multimodal undertaking of text combined with speech and vision. Lastly, interpretability, fairness, privacy-by-design and continuous learning must be worked on so that we can have trustful and long-lived deployments.

VII. Conclusion

The paper presents a single unified model of integrating emotion-conditional text generation with transformer-based classification to resolve imbalance and heterogeneity in emotion data sets. The method attaining extensive improvements on GoEmotions, TweetEval, and DAIR Emotion by synthesizing label-faithful samples representing sparse emotions and hyperparameter refinement in RoBERTa using Optuna, the method employs automated hyperparameter search. However, these minority categories are stronger at improving despite being spread widely across the board and hence, the decision boundaries are calibrated and there is more robust generalization. The pipeline is succinct on small hardware, manifest on the limiting, judicious consumption of generative AI on commonplace workflows and recorded in the same format of end-to-end disclosures in accompanying notebooks. In the future, they may extend to multilingual and multimodal, domain adaptation, and continuous learning.

VIII. Acknowledgments and CSRAI Statement

The creators of GoEmotions, TweetEval, and DAIR Emotion have made their datasets available, and the open-source community made SmolLM-135M-Instruct and RoBERTa-base. We realize that Google Colab provides us with computational resources. Disclosure: the authors applied Gemini on Google Colab in a small assistive capacity to scaffold and refactor its code, write augmentation prompts, and correct the grammar and diction. The authors have made all modeling choices, curation process of data, results, and analyses themselves. Any AI-recommended material was checked, edited, and confirmed prior to being added and no trade secrets or personal information was submitted to the tool. The complete implementation is available.

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